D.4. <future> header

The `<future>` header provides facilities for handling asynchronous results from operations that may be performed on another thread.

Header contents

```
namespace std
{
    enum class future_status {
        ready, timeout, deferred };

    enum class future_errc
    {

         broken_promise,
         future_already_retrieved,
        promise_already_satisfied,
        no_state
     };

     class future_error;

     const error_category& future_category();

     error_code make_error_code(future_errc e);
     error_condition make_error_condition(future_errc e);

     template<typename ResultType>
     class future;

     template<typename ResultType>
     class shared_future;

     template<typename ResultType>
     class promise;

     template<typename FunctionSignature>
     class packaged_task; // no definition provided

     template<typename ResultType,typename ... Args>
     class packaged_task<ResultType (Args...)>;

     enum class launch {
         async, deferred
     };
     template<typename FunctionType,typename ... Args>
     future<result_of<FunctionType(Args...)>::type>
```